Grade 9 Practice

Standard A.SSE.1

Name: _____ Date: _____

Computation

1. In $3x + 7$, what is the coefficient of x? _____

2. In $2x^2 - 5$, what is the constant term? _____

3. How many terms are in $4x - 9$? _____

4. In $5(x + 2)$, what is the factor outside the parentheses? _____

Word problems

5. The cost is $C = 12h + 30$, where h is hours. What does 12 represent?

Show your work and answer.

6. The value is $V = 500 - 25t$, where t is days. What does 25 represent?

Show your work and answer.

Grade 9 Practice

Standard A.SSE.2

Name: _____ Date: _____

Computation

1. Rewrite $x^2 - 9$ as a product of two factors. _____

2. Factor $x^2 + 5x + 6$. _____

3. Factor $4x^2 - 1$. _____

4. Rewrite $a^2 + 2ab + b^2$ as a square of a binomial. _____

Word problems

5. A square patio has area $x^2 + 6x + 9$. What is the side length?

Show your work and answer.

6. Factor $9 - x^2$.

Show your work and answer.

Grade 9 Practice

Standard A.APR.1

Name: _____ Date: _____

Computation

1. Simplify $(x + 3) + (2x - 1)$. _____

2. Multiply $(x + 2)(x + 5)$. _____

3. Simplify $(3x - 4) - (x + 6)$. _____

4. Multiply $2x(x - 3)$. _____

Word problems

5. A rectangle has sides $x + 2$ and $x + 4$. What is its area?

Show your work and answer.

6. Add $(2x^2 + 3)$ and $(x^2 - x)$. What is the sum?

Show your work and answer.

Grade 9 Practice

Standard A.CED.1

Name: _____ Date: _____

Computation

1. Write an equation: 5 more than 3 times n is 20. _____

2. Write an equation: twice x minus 7 equals 11. _____

3. Write an inequality: y is at least 4. _____

4. Write an equation: x divided by 4 equals 6. _____

Word problems

5. A phone plan costs $15 plus $0.10 per text. Your bill is $25. Write an equation using t for texts.

Show your work and answer.

6. A team scored 3 times its points p plus 4 points, for a total of 31. Write an equation.

Show your work and answer.

Grade 9 Practice

Standard A.CED.2

Name: _____ Date: _____

Computation

1. In the line y = 2x + 1, what is the y-intercept? _____

2. In the line y = -x + 4, what is the slope? _____

3. Is the point (2, 5) on the line y = 2x + 1? Write yes or no.

4. Is the point (0, 3) on the line y = x - 3? Write yes or no.

Word problems

5. A cost is y = 5x + 10, where x is hours. What does the 10 represent?

Show your work and answer.

6. A car's distance is y = 60x, where x is hours. What does the slope represent?

Show your work and answer.

Grade 9 Practice

Standard A.CED.3

Name: _____ Date: _____

Computation

1. Solve the system: x + y = 10 and x - y = 4. Write (x, y).

2. Solve the system: y = 2x and y = x + 3. Write (x, y).

3. Pens cost p and pencils cost c. 3p + 2c = 11 and p + 2c = 7. What is the cost of one pen? _____

4. Adult tickets a and child tickets c: a + c = 50 and a - c = 10. What is a?

_____

Word problems

5. Tickets: a + c = 100 and 8a + 5c = 620, where a is adult and c is child tickets. How many of each were sold?

Show your work and answer.

6. Nickels n and dimes d: n + d = 12 and 5n + 10d = 85. How many nickels and how many dimes?

Show your work and answer.

Grade 9 Practice

Standard A.CED.4

Name: _____ Date: _____

Computation

1. Solve $y = mx + b$ for b. b = _____

2. Solve $2x + y = 6$ for y. y = _____

3. Solve $A = lw$ for w. w = _____

4. Solve $C = 2\pi r$ for r. r = _____

Word problems

5. Distance $d = rt$. Solve for t.

Show your work and answer.

6. Perimeter $P = 2l + 2w$. Solve for w.

Show your work and answer.

Grade 9 Practice

Standard A.REI.1

Name: _____ Date: _____

Computation

1. Solve x/2 = 5. x = _____

2. Solve 4x - 8 = 0. x = _____

3. Solve 5 = x - 3. x = _____

4. Solve 3x = -9. x = _____

Word problems

5. Sam says x = 2 solves 3x + 1 = 7. Is he right? Write yes or no.

Show your work and answer.

6. Mia solves 2x + 3 = 11 by subtracting 3 and then dividing by 2. Does she get x = 4? Write yes or no.

Show your work and answer.

Grade 9 Practice

Standard A.REI.3

Name: _____ Date: _____

Computation

1. Solve $x + 3 > 7$. $x >$ _____

2. Solve $2x \leq 10$. $x \leq$ _____

3. Solve $-3x < 9$. $x >$ _____

4. Solve $4x - 5 \geq 11$. $x \geq$ _____

Word problems

5. Tickets cost $8 each. You can spend at most $64. Write an inequality for t tickets and solve it.

Show your work and answer.

6. A game needs at least 30 points. You earn 4 points per round. What is the fewest whole number of rounds needed?

Show your work and answer.

Grade 9 Practice

Standard A.REI.4

Name: _____ Date: _____

Computation

1. Solve $x^2 = 16$. $x =$ _____

2. Solve $x^2 - 9 = 0$. $x =$ _____

3. Solve $(x - 2)^2 = 25$. $x =$ _____

4. Solve $x^2 = 0$. $x =$ _____

Word problems

5. A square has area 49 square units. What is its side length?

Show your work and answer.

6. A square has area 81 square units. What is its side length?

Show your work and answer.

Grade 9 Practice

Standard A.REI.5

Name: _____ Date: _____

Computation

1. Does multiplying x + y = 4 by 2 give 2x + 2y = 8? Write yes or no.

2. Is 2x = 6 equivalent to x = 3? Write yes or no.

3. Multiply x + y = 4 by 2. Write the new equation.

4. Is 3x = 6 equivalent to 6x = 12? Write yes or no.

Word problems

5. Solve 2x - 4 = 6 and 4x - 8 = 12. Do they have the same solution? Write yes or no.

Show your work and answer.

6. If you multiply both sides of an equation by 0, do the solutions stay the same? Write yes or no.

Show your work and answer.

Grade 9 Practice

Standard A.REI.6

Name: _____ Date: _____

Computation

1. Solve the system: y = x + 2 and y = -x. Write (x, y).

2. Solve the system: y = 2x + 1 and y = 2x + 4. Write yes if there is a solution or no if there is none.

3. Solve the system: x + y = 6 and x - y = 2. Write (x, y).

4. Solve the system: y = 3x and y = x + 4. Write (x, y).

Word problems

5. Plan A costs y = 5x + 10 and plan B costs y = 3x + 30. For what x do the costs match?

Show your work and answer.

6. Solve the system: x + y = 20 and y = 3x. Write (x, y).

Show your work and answer.

Grade 9 Practice

Standard F.IF.1

Name: _____ Date: _____

Computation

1. If $f(x) = 2x + 3$, what is $f(4)$? _____

2. If $g(x) = x^2$, what is $g(-3)$? _____

3. If $h(x) = x - 5$, what is $h(10)$? _____

4. If $f(x) = 7x - 2$, what is $f(0)$? _____

Word problems

5. A phone cost is $c(t) = 10 + 5t$, where t is months. What is $c(3)$?

Show your work and answer.

6. If $f(x) = x^2 - 1$, which input gives $f(x) = 0$? Write your answer.

Show your work and answer.

Grade 9 Practice

Standard F.IF.2

Name: _____ Date: _____

Computation

1. If $f(x) = 3x + 1$, find $f(2)$. _____

2. If $g(x) = x^2 + 2x$, find $g(3)$. _____

3. If $h(x) = 10 - x$, find $h(-4)$. _____

4. If $f(x) = x/2$, find $f(8)$. _____

Word problems

5. The temperature is $T(h) = 30 - 2h$, where h is hours. What is $T(5)$?

Show your work and answer.

6. If $A(s) = s^2$, find $A(6)$.

Show your work and answer.

Grade 9 Practice

Standard F.IF.4

Name: _____ Date: _____

Computation

1. What is the y-intercept of y = 3x - 6? _____

2. What is the x-intercept of y = 2x - 8? _____

3. What is the slope of y = -x + 2? _____

4. Does the graph of y = 2x increase? Write yes or no.

Word problems

5. The graph of $y = x^2$ has its lowest point at (0, 0). Is that the vertex? Write yes or no.

Show your work and answer.

6. A temperature graph falls after noon. Is it increasing or decreasing after noon?

Show your work and answer.

Grade 9 Practice

Standard F.BF.1

Name: _____ Date: _____

Computation

1. A pile starts at 10 and adds 4 each day. Write the function f(x) for x days.

2. A number is multiplied by 3. Write f(x).

3. A number is squared and then 1 is subtracted. Write f(x).

4. A number is doubled and then 5 is added. Write f(x).

Word problems

5. A phone plan costs $30 plus $5 per month. Write a function f(m) for the cost after m months.

Show your work and answer.

6. A bacteria count starts at 100 and doubles every hour. Write a function f(t) for t hours.

Show your work and answer.

Grade 9 Practice

Standard F.LE.1

Name: _____ Date: _____

Computation

1. A table increases by 2 at each step. Is it linear or exponential?

2. A table doubles at each step. Is it linear or exponential?

3. A table increases by 5 at each step. Is it linear or exponential?

4. A table multiplies by 3 at each step. Is it linear or exponential?

Word problems

5. A population grows by 10% each year. Is it linear or exponential?

Show your work and answer.

6. A salary increases by $500 each year. Is it linear or exponential?

Show your work and answer.

Grade 9 Practice

Standard F.LE.2

Name: _____ Date: _____

Computation

1. A line passes through (0, 3) and (1, 5). Write the equation.

2. An exponential function passes through (0, 2) and (1, 6). Write the equation.

3. A line has slope 4 and y-intercept -1. Write the equation.

4. An exponential function starts at 5 and doubles each step. Write the equation.

Word problems

5. Savings start at $100 and grow by $20 each month. Write an equation for y after x months.

Show your work and answer.

6. A value is cut in half each year, starting at 800. Write an equation for y after x years.

Show your work and answer.

Grade 9 Practice

Standard S.ID.1

Name: _____ Date: _____

Computation

1. Find the range of 2, 4, 6, and 8. _____

2. Find the mean of 1, 2, 3, 4, and 5. _____

3. Find the median of 1, 3, 5, and 7. _____

4. Find the mode of 2, 2, 3, and 4. _____

Word problems

5. Five scores are 80, 85, 90, 95, and 100. What is the mean?

Show your work and answer.

6. A histogram has 6 values in the 10–20 bin and 4 in the 20–30 bin. How many values in all?

Show your work and answer.

Grade 9 Practice

Standard S.ID.2

Name: _____ Date: _____

Computation

1. Which measure of center is resistant to outliers? Write mean or median.

2. Find the mean of the data 1, 2, 3, 4, and 5. _____

3. What does standard deviation measure? Write spread or center.

4. For symmetric data, how do the mean and median compare? Write equal or not equal.

Word problems

5. Income data is skewed to the right. Which measure of center is better to use?

Show your work and answer.

6. Data are 10, 12, 14, and 16. What is the mean?

Show your work and answer.